

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#04)

Use of Venus Simulator for Register Level Execution of RISC-V Arithmetic and Memory Instructions

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MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____

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Version: 1.0

Task 1: Complete following tasks in Lab.

Copy your Venus Editor code, paste it into your submission copy and upload on GCR. Also translate to machine language representation (if required).

1. Write RISC-V assembly code for given C program:

Consider A=5, B=4

i- C = A+B

ii- D = A-B

iii- E = A*B

iv- F = A^2

v- Separate the different fields of these RISC-V instructions according to their format and write decimal values. That is, write the machine language representation (expressed in decimal).

Description:

In this task venus editor was used to write assembly of risc-V and different operations were performed such as add,sub,mul,and power .

Code:

```
.text
.globl main
main:
    addi s4, x0, 5
    addi s5, x0, 4
    add s6, s4, s5
    sub s7, s4, s5
```

mul s8, s4, s5

mul s9, s4, s4

ret

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s4 (x20)

5

s5 (x21)

0

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s4 (x20)	5
s5 (x21)	4
s6 (x22)	0
s7 (x23)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s4 (x20)	5
s5 (x21)	4
s6 (x22)	9
s7 (x23)	0
s8 (x24)	0
s9 (x25)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s5 (x21)	4
s6 (x22)	9
s7 (x23)	1
s8 (x24)	0
s9 (x25)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s6 (x22)	9
s7 (x23)	1
s8 (x24)	20
s9 (x25)	0
s10 (x26)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00500A13	addi x20 x0 5	addi s4, x0, 5
0x4	0x00400A93	addi x21 x0 4	addi s5, x0, 4
0x8	0x015A0B33	add x22 x20 x21	add s6, s4, s5
0xc	0x415A0BB3	sub x23 x20 x21	sub s7, s4, s5
0x10	0x035A0C33	mul x24 x20 x21	mul s8, s4, s5
0x14	0x034A0CB3	mul x25 x20 x20	mul s9, s4, s4
0x18	0x00008067	jalr x0 x1 0	ret

s4 (x20)	5
s5 (x21)	4
s6 (x22)	9
s7 (x23)	1
s8 (x24)	20
s9 (x25)	25
s10 (x26)	0
s11 (x27)	0

Task 2: Convert the following C code into RISC-V assembly language. Use appropriate registers for the variables.

$m = (a - c) + (d + e) * f$

Description:

In this task venus editor was used to write assembly of risc-V and different operations were performed such as add,sub,mul, and result is stored in m.

Code:

```
.text
.globl main
main:
    addi s10, x0, 10
    addi s11, x0, 20
    addi t3, x0, 30
    addi t4, x0, 15
    addi t5, x0, 68
    sub t6, s10, s11
    add a0, t3, t4
    mul a1, a0, t5
    add a2, t6, a1
    ret
```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalt x0 x1 0	ret

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret
PC Machine Code Basic Code Original Code			
s8 (x24) 0			
s9 (x25) 0			
s10 (x26) 10			
s11 (x27) 0			
t3 (x28) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret

s9 (x25)	0
s10 (x26)	10
s11 (x27)	20
t3 (x28)	0
t4 (x29)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret

s9 (x25)	0
s10 (x26)	10
s11 (x27)	20
t3 (x28)	30
t4 (x29)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret
s8 (x24) 0			
s9 (x25) 0			
s10 (x26) 10			
s11 (x27) 20			
t3 (x28) 30			
t4 (x29) 15			
t5 (x30) 0			
t6 (x31) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret

s10 (x26)

s11 (x27)

t3 (x28)

t4 (x29)

t5 (x30)

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret

s10 (x26)	10
s11 (x27)	20
t3 (x28)	30
t4 (x29)	15
t5 (x30)	68
t6 (x31)	-10

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E50533	mul x10 x10 x30	mul a0, a0, t5
0x20	0x00AF85B3	add x11 x31 x10	add a1, t6, a0
0x24	0x00008067	jalr x0 x1 0	ret

s1 (x9)

a0 (x10)

a1 (x11)

a2 (x12)

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E505B3	mul x11 x10 x30	mul a1, a0, t5
0x20	0x00BF8633	add x12 x31 x11	add a2, t6, a1
0x24	0x00008067	jalr x0 x1 0	ret

t2 (x7)	0		
s0 (x8)	0		
s1 (x9)	0		
a0 (x10)	45		
a1 (x11)	3060		
a2 (x12)	0		

PC	Machine Code	Basic Code	Original Code
0x0	0x00A00D13	addi x26 x0 10	addi s10, x0, 10
0x4	0x01400D93	addi x27 x0 20	addi s11, x0, 20
0x8	0x01E00E13	addi x28 x0 30	addi t3, x0, 30
0xc	0x00F00E93	addi x29 x0 15	addi t4, x0, 15
0x10	0x04400F13	addi x30 x0 68	addi t5, x0, 68
0x14	0x41BD0FB3	sub x31 x26 x27	sub t6, s10, s11
0x18	0x01DE0533	add x10 x28 x29	add a0, t3, t4
0x1c	0x03E505B3	mul x11 x10 x30	mul a1, a0, t5
0x20	0x00BF8633	add x12 x31 x11	add a2, t6, a1
0x24	0x00008067	jalr x0 x1 0	ret

s1 (x9)	0
a0 (x10)	45
a1 (x11)	3060
a2 (x12)	3050
a3 (x13)	0

Task 3: Write RISC-V assembly language program to perform AND, OR, XOR on the following numbers using Venus.

0x f f f

0x f 0 f

Description:
In this task venus editor was used to write assembly of risc-V and different operations were performed such as AND,OR,XOR using hex inputs.
Code:
li t0, 0xFFFF li t1, 0xF0F and t2, t0, t1 or t3, t0, t1 xor t4, t0, t1
Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1
PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1
tp (x4) 0			
t0 (x5) 4096			
t1 (x6) 0			
t2 (x7) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1

tp (x4)

t0 (x5)

t1 (x6)

t2 (x7)

s0 (x8)

PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1

PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1
cp (x4) 0 t0 (x5) 4095 t1 (x6) 3855 t2 (x7) 3855 s0 (x8) 0			
PC	Machine Code	Basic Code	Original Code
0x0	0x000012B7	lui x5 1	li t0, 0xFFFF
0x4	0xFFFF28293	addi x5 x5 -1	li t0, 0xFFFF
0x8	0x00001337	lui x6 1	li t1, 0xF0F
0xc	0xF0F30313	addi x6 x6 -241	li t1, 0xF0F
0x10	0x0062F3B3	and x7 x5 x6	and t2, t0, t1
0x14	0x0062EE33	or x28 x5 x6	or t3, t0, t1
0x18	0x0062CEB3	xor x29 x5 x6	xor t4, t0, t1

s10 (x26)	0
s11 (x27)	0
t3 (x28)	4095
t4 (x29)	0

Task 4: The below RISC-V assembly code is supposedly an assembled version of the C code, also shown below. Use Venus to run and observe the registers values in Venus register view to verify the code is indeed running as expected. In case of syntax or logical errors, fix the assembly code until it works as intended.

#Assembly code

.section .text

.global arraysum

arraysum:

a0 = int a[]

a1 = int size

t0 = ret

t1 = i

li t0, 0 # ret = 0

li t1, 0 # i = 0

1: # For loop

bge t1, a1, 1f # if i >= size, break

slli t2, t1, 2 # Multiply i by 4 (1 << 2 = 4)

add t2, a0, t2 # Update memory address

lw t2, 0(t2) # Dereference address to get integer

add t0, t0, t2 # Add integer value to ret

```

addi t1, t1, 1 # Increment the iterator
j 1b # Jump back to start of loop (1 backwards)
1:
mv a0, t0 # Move t0 (ret) into a0
ret # Return via return address register
#HLL code (Sum of an Integer Array)
int arraysum(int a[], int size) {
int ret = 0;
int i;
for (i = 0; i < size; i++) {
ret = ret + a[i];
}
return ret;
}

```

Description:

In this task venus editor was used to write assembly of risc-V . Already given code is tested and errors were fixed and required results were obtained.

Code:

```

.data
array: .word 1, 2, 3, 4, 5
.text
main:
    la a0, array
    li a1, 5
    jal ra, arraysum
    li a7, 10
    ecall
arraysum:
    li t0, 0
    li t1, 0
1:
    bge t1, a1, 1f
    slli t2, t1, 2
    add t2, a0, t2
    lw t2, 0(t2)
    add t0, t0, t2
    addi t1, t1, 1
    j 1b
1:
    mv a0, t0
    ret

```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x10000517	auipc x10 65536	la a0, array
0x4	0x00050513	addi x10 x10 0	la a0, array
0x8	0x00500593	addi x11 x0 5	li a1, 5
0xc	0x00C000EF	jal x1 12	jal ra, arraysum
0x10	0x00A00893	addi x17 x0 10	li a7, 10
0x14	0x00000073	ecall	ecall
0x18	0x00000293	addi x5 x0 0	li t0, 0
0x1c	0x00000313	addi x6 x0 0	li t1, 0
0x20	0x00B35063	bge x6 x11 0	bge t1, a1, 1f
0x24	0x00231393	slli x7 x6 2	slli t2, t1, 2
0x28	0x007503B3	add x7 x10 x7	add t2, a0, t2
0x2c	0x0003A383	lw x7 0(x7)	lw t2, 0(t2)

ra (x1)	0
sp (x2)	2147483632
gp (x3)	268435456
tp (x4)	0
t0 (x5)	0
t1 (x6)	0
t2 (x7)	0
s0 (x8)	0
s1 (x9)	0
a0 (x10)	268435456

PC	Machine Code	Basic Code	Original Code
0x0	0x10000517	auipc x10 65536	la a0, array
0x4	0x00050513	addi x10 x10 0	la a0, array
0x8	0x00500593	addi x11 x0 5	li a1, 5
0xc	0x00C000EF	jal x1 12	jal ra, arraysum
0x10	0x00A00893	addi x17 x0 10	li a7, 10
0x14	0x00000073	ecall	ecall
0x18	0x00000293	addi x5 x0 0	li t0, 0
0x1c	0x00000313	addi x6 x0 0	li t1, 0
0x20	0x00B35063	bge x6 x11 0	bge t1, a1, 1f
0x24	0x00231393	slli x7 x6 2	slli t2, t1, 2
0x28	0x007503B3	add x7 x10 x7	add t2, a0, t2
0x2c	0x0003A383	lw x7 0(x7)	lw t2, 0(t2)

s1 (x9)	0
a0 (x10)	268435456
a1 (x11)	5
a2 (x12)	0

PC	Machine Code	Basic Code	Original Code
0x0	0x10000517	auipc x10 65536	la a0, array
0x4	0x00050513	addi x10 x10 0	la a0, array
0x8	0x00500593	addi x11 x0 5	li a1, 5
0xc	0x00C000EF	jal x1 12	jal ra, arraysum
0x10	0x00A00893	addi x17 x0 10	li a7, 10
0x14	0x00000073	ecall	ecall
0x18	0x00000293	addi x5 x0 0	li t0, 0
0x1c	0x00000313	addi x6 x0 0	li t1, 0
0x20	0x00B35063	bge x6 x11 0	bge t1, a1, 1f
0x24	0x00231393	slli x7 x6 2	slli t2, t1, 2
0x28	0x007503B3	add x7 x10 x7	add t2, a0, t2
0x2c	0x0003A383	lw x7 0(x7)	lw t2, 0(t2)

t0 (x5)

t1 (x6)

t2 (x7)

s0 (x8)

s1 (x9)

a0 (x10)

Task 5: Write RISC-V assembly code to write table of two. You need to initialize table of two in the data section using .data in Venus. After this, load table values of two into different registers using lw command.

Description:
In this task venus editor was used to write assembly of risc-V . Certain libraries are used such as .data, .word and using lw command values are stored in registers
Code:
<pre>.data table: .word 3, 6, 9, 12, 15, 18, 21, 24, 27, 30 .text main: la t0, table lw t1, 0(t0) lw t2, 4(t0) lw t3, 8(t0) lw t4, 12(t0) lw t5, 16(t0) lw t6, 20(t0) lw a0, 24(t0) lw a1, 28(t0) lw a2, 32(t0) lw a3, 36(t0)</pre>
Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)
tp (x4) 0			
t0 (x5) 268435456			
t1 (x6) 0			
t2 (x7) 0			
s0 (x8) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)

tp (x4)	0
t0 (x5)	268435456
t1 (x6)	3

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)
tp (x4) 0			
t0 (x5) 268435456			
t1 (x6) 3			
t2 (x7) 6			
s0 (x8) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)
t3 (x28) 9			
t4 (x29) 12			
t5 (x30) 15			

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)
s11 (x27) 0 t3 (x28) 9 t4 (x29) 12 t5 (x30) 15 t6 (x31) 18 s1 (x9) 0 a0 (x10) 21 a1 (x11) 0			

PC	Machine Code	Basic Code	Original Code
0x0	0x10000297	auipc x5 65536	la t0, table
0x4	0x00028293	addi x5 x5 0	la t0, table
0x8	0x0002A303	lw x6 0(x5)	lw t1, 0(t0)
0xc	0x0042A383	lw x7 4(x5)	lw t2, 4(t0)
0x10	0x0082AE03	lw x28 8(x5)	lw t3, 8(t0)
0x14	0x00C2AE83	lw x29 12(x5)	lw t4, 12(t0)
0x18	0x0102AF03	lw x30 16(x5)	lw t5, 16(t0)
0x1c	0x0142AF83	lw x31 20(x5)	lw t6, 20(t0)
0x20	0x0182A503	lw x10 24(x5)	lw a0, 24(t0)
0x24	0x01C2A583	lw x11 28(x5)	lw a1, 28(t0)
0x28	0x0202A603	lw x12 32(x5)	lw a2, 32(t0)
0x2c	0x0242A683	lw x13 36(x5)	lw a3, 36(t0)

a0 (x10)	21
a1 (x11)	24
a2 (x12)	27